SAT Solver Problem Set and Demonstration

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1 Benchmark Problem

2 Official Source

The link to benchmark problem is as follows:

`https://users.aalto.fi/~tjunttil/2022-DP-AUT/notes-sat/solving.html`

```
p cnf 729 3270
```

3 Self-Made Problem

```
p cnf 3 4
1 2 0
-1 3 0
-2 3 0
-3 0
```

4 CNF Formula in Logical Form

$$(A \vee B)$$
$$(\neg A \vee C)$$
$$(\neg B \vee C)$$
$$(\neg C)$$

5 Explanation of the Syntax (DIMACS Format)

DIMACS is the standard format used for CNF encoding in SAT solvers.

Assume we have three variables:

$$A, B, C$$

And the propositions:

$$A \vee B$$
$$B \vee \neg C$$
$$C$$

We assign numbers:

$$A = 1$$
$$B = 2$$
$$C = 3$$

Rules:

- Positive literal → positive number
- Negative literal → negative number
- Space separates literals (disjunction)
- New line separates clauses
- End each clause with 0
- Header format: p cnf ¡variables¿ ¡clauses¿

Example:

```
p cnf 3 3
1 2 0
2 -3 0
3 0
```

6 Demonstration of the Example Run

Ensure the SAT solver is successfully installed.

6.1 Step 1: Confirm Installation

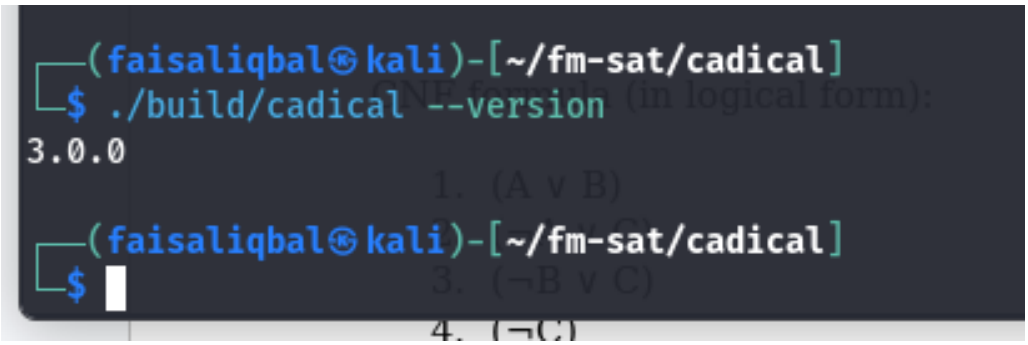


Figure 1: Checking Installation

6.2 Step 2: Create CNF File

Create the CNF file inside the solver directory.

6.3 Step 3: Confirm File Creation

Use ls command to verify.

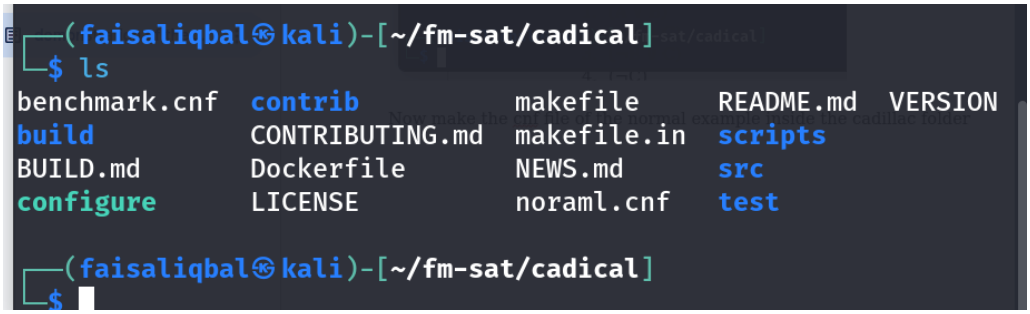


Figure 2: Confirming File Exists

6.4 Step 4: Run the Solver

Command:

```
./build/cadical fileName.cnf
```

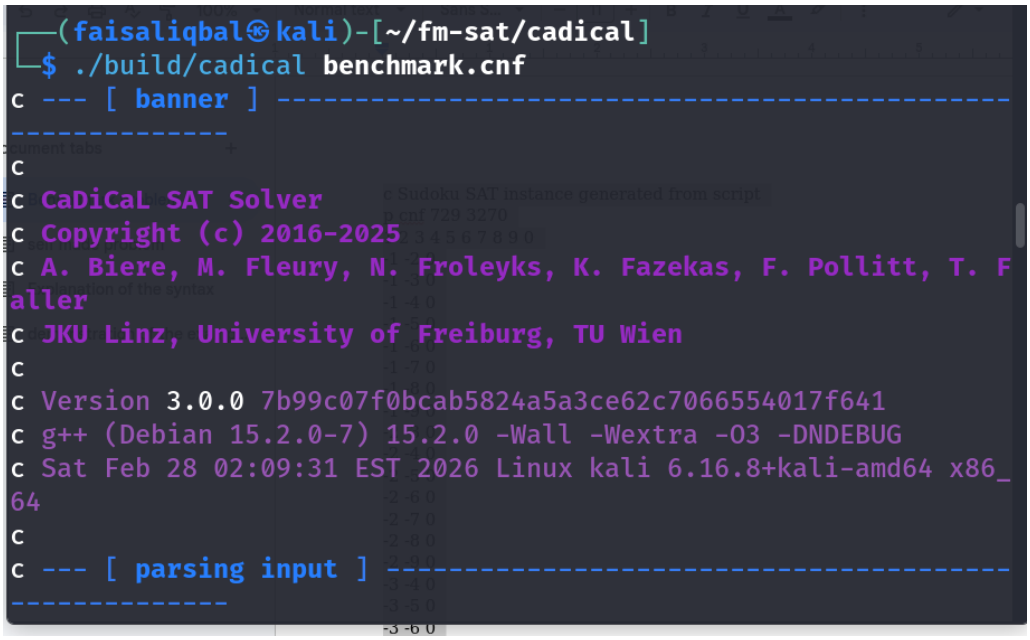


Figure 3: Running the Solver

6.5 Solver Output Stages

6.5.1 Parsing Stage

```
c
c time measured in process time since initialization
c
c  seconds  reduction conflicts       size tier2     variables
c          MB    restarts   redundant    glue  trail      remaining
c           level     rate       size/glu tier1 irredundant
c
c *  0.01  5  0 0   0  0    0    0 0.0 0 0 2 6  0% 3240 459 63%
c (  0.01  5  0 0   0  0    0    0 0.0 0 0 2 6  0% 3240 459 63%
c .  0.01  5  0 0   0  0    0    0 0.0 0 0 2 6  0% 2040 459 63%
c f  0.02  6  0 0   0  0    0    0 0.0 0 0 2 6  0% 1275 612 84%
c e  0.03  6  0 0   0  0    0    0 0.0 0 0 2 6  0% 1144 461 63%
c s  0.03  6  0 0   0  0    0    0 0.0 0 0 2 6  0% 1144 461 63%
c P  0.03  6  0 0   0  0    0    0 0.0 0 0 2 6  0% 1144 461 63%
c )  0.03  6  0 0   0  0    0    0 0.0 0 0 2 6  0% 1144 461 63%
c {  0.03  6  0 0   0  0    0    0 0.0 0 0 2 6  0% 1144 461 63%
c i  0.03  6  9 0   0  2   64   29 1.6 4 3 2 6 37% 1144 460 63%
c i  0.03  6 23 0  10  3  166   98 1.5 5 3 2 6 43% 1144 457 63%
c -  0.03  6 27 1  33  4  300  174 1.5 6 4 2 6 45% 1130 457 63%
c -  0.04  6 17 2  71  3  735  467 1.6 7 4 2 6 42% 1130 457 63%
c i  0.04  6 17 2  77  3  766  495 1.6 7 4 2 6 42% 1130 456 63%
c i  0.04  6 17 2  77  3  934  621 1.6 7 4 2 6 45% 1130 454 62%
c i  0.04  6 17 2  77  3  987  660 1.6 7 4 2 6 45% 1130 453 62%
c }  0.04  6 17 2  77  2 1001  673 1.6 7 4 2 6 46% 1130 453 62%
c [  0.04  6  0 2  77  0 1001  673 0.0 0 0 2 6  0% 1130 453 62%
c O  0.04  6 15 2  77 16 1002  674 1.2 5 4 2 6 38% 1130 453 62%
c d  0.04  6 19 2  77  1 1233  847 2.1 9 4 2 6 45% 1117 452 62%
c
```

```
c
c  seconds  reduction conflicts       size tier2     variables
c          MB    restarts   redundant    glue  trail      remaining
c           level     rate       size/glu tier1 irredundant
c
c t  0.04  6 19 2  77  1 1233  846 2.1 9 4 2 6 45% 1117 452 62%
c -  0.05  6 13 3  77  1 1413  845 1.9 8 4 2 6 44% 1117 452 62%
c i  0.05  6 11 3  77  1 1625  953 1.7 8 5 2 6 43% 1117 451 62%
c ]  0.05  6 12 3  77  1 1774 1054 1.7 8 5 2 6 43% 1117 451 62%
c {  0.05  6 17 3  77  2 1774 1054 1.6 7 4 2 6 46% 1117 451 62%
c i  0.05  6 17 3  79  2 1967 1229 1.8 7 4 2 6 48% 1117 449 62%
c i  0.05  6 17 3  79  2 2048 1287 1.8 7 4 2 6 48% 1117 448 61%
c i  0.05  6 17 3  79  2 2068 1304 1.8 7 4 2 6 48% 1117 447 61%
c i  0.06  6 15 3  79  2 2280 1432 1.8 7 4 2 6 48% 1117 446 61%
c i  0.06  6 15 3  79  2 2314 1225 1.8 7 4 2 6 39% 1099 444 61%
c -  0.06  6 15 4  79  2 2355 1090 1.8 7 4 2 6 42% 1093 444 61%
c }  0.06  6 15 4  79  2 2379 1113 1.8 7 4 2 6 47% 1093 444 61%
c 1  0.06  6 15 4  79  2 2379 1113 1.8 7 4 2 6 47% 1093 444 61%
c
```

Figure 4: Parsing Started

6.5.2 Solving Process and Statistics

```
c --- [ result ] ------------------------------------------------------------
c
s SATISFIABLE
v -1 -2 -3 -4 5 -6 -7 -8 -9 -10 -11 12 -13 -14 -15 -16 -17 -18 -19 -20 -21 22
v -23 -24 -25 -26 -27 -28 -29 -30 -31 -32 33 -34 -35 -36 -37 -38 -39 -40 -41
v -42 43 -44 -45 -46 -47 -48 -49 -50 -51 -52 53 -54 -55 -56 -57 -58 -59 -60
v -61 -62 63 64 -65 -66 -67 -68 -69 -70 -71 -72 -73 74 -75 -76 -77 -78 -79 -80
v -81 -82 -83 -84 -85 -86 87 -88 -89 -90 -91 -92 -93 -94 -95 -96 97 -98 -99
v -100 101 -102 -103 -104 -105 -106 -107 -108 109 -110 -111 -112 -113 -114
v -115 -116 -117 -118 -119 -120 -121 -122 -123 -124 -125 126 -127 -128 -129
v -130 131 -132 -133 -134 -135 -136 -137 138 -139 -140 -141 -142 -143 -144
v -145 -146 -147 148 -149 -150 -151 -152 -153 -154 -155 -156 -157 -158 -159
v -160 161 -162 163 -164 -165 -166 -167 -168 -169 -170 -171 -172 -173 -174
v -175 -176 -177 -178 -179 180 -181 -182 -183 -184 -185 -186 -187 188 -189
v -190 -191 192 -193 -194 -195 -196 -197 -198 -199 -200 -201 202 -203 -204
v -205 -206 -207 -208 209 -210 -211 -212 -213 -214 -215 -216 -217 -218 -219
v -220 221 -222 -223 -224 -225 -226 -227 -228 -229 -230 231 -232 -233 -234
v -235 -236 -237 -238 -239 -240 241 -242 -243 -244 -245 -246 -247 -248 -249
v -250 251 -252 -253 -254 -255 -256 257 -258 -259 -260 -261 -262 -263 -264
v -265 -266 -267 -268 -269 270 -271 -272 -273 -274 -275 -276 277 -278 -279
v -280 -281 -282 -283 -284 285 -286 -287 -288 289 -290 -291 -292 -293 -294
v -295 -296 -297 -298 -299 -300 301 -302 -303 -304 -305 -306 -307 308 -309
v -310 -311 -312 -313 -314 -315 -316 -317 318 -319 -320 -321 -322 -323 -324
v -325 -326 -327 328 -329 -330 -331 -332 -333 -334 335 -336 -337 -338 -339
v -340 -341 -342 -343 -344 -345 -346 -347 348 -349 -350 -351 -352 -353 -354
v -355 -356 -357 -358 359 -360 -361 -362 -363 -364 365 -366 -367 -368 -369
v -370 -371 372 -373 -374 -375 -376 -377 -378 -379 -380 -381 -382 -383 -384
v 385 -386 -387 -388 -389 -390 -391 -392 -393 -394 -395 396 397 -398 -399 -400
```

Figure 5: Solving and Conflict Statistics

6.5.3 Final Results

```
c --- [ run-time profiling ] ---------------------------------------------
c
c process time taken by individual solving procedures
c (percentage relative to process time for solving)
c
c         0.03   59.96% search
c         0.02   39.06% unstable
c         0.02   33.94% preprocess
c         0.02   33.46% simplify
c         0.01   18.52% parse
c         0.01   18.18% stable
c         0.01   14.15% sweep
c         0.00    5.44% fastelim
c         0.00    4.29% congruence
c         0.00    3.61% factor
c         0.00    3.16% subsume
c         0.00    1.38% lucky
c         0.00    0.87% backbone
c         0.00    0.45% probe
c   ================================
c         0.05   80.55% solve
c
c last line shows process time for solving
c (percentage relative to total process time)
```

Figure 6: Runtime profiling

7 Interpreting the Sudoku Output

The first 9 values correspond to the first cell, the next 9 to the second cell, and so on.

A cell takes the value equal to:

$$(\text{positive literal mod } 9)$$

Example result:

$$5 \quad 3 \quad 4$$

for the first three cells.

8 Performance Statistics

8.1 Time Breakdown

```
c --- [ statistics ] ---------------------------------------------------
c
c chronological:                728        30.60 %  of conflicts
c conflicts:                   2379     49095.07    per second
c backbone:                    6524       739.68 %  of vars
c decisions:                   4529     93464.31    per second
c eliminated:                   151        17.12 %  of all variables
c factored:                     153        34.23 %  of variables
c fixed:                        286        32.43 %  of all variables
c learned:                     2303        96.81 %  per conflict
c learned_lits:               27959       100.00 %  learned literals
c minimized:                      0         0.00 %  learned literals
c shrunken:                    9043        32.34 %  learned literals
c minishrunken:                1949         6.97 %  learned literals
c otfs:                          39         1.64 %  of conflict
c propagations:               75721         1.56 M  per second
c reduced:                      285        11.98 %  per conflict
c rephased:                       1      2379.00    interval
c restarts:                      79        30.11    interval
c stabilizing:                    1        32.20 %  of conflicts
c substituted:                    1         0.11 %  of all variables
c subsumed:                     637         8.81 %  of all clauses
c strengthened:                 153         2.12 %  of all clauses
c ticks:                     704277         9.76    propagation
c weakened:                     712         4.33    average size
c congruence:                     1      2379.00    interval
c
c seconds are measured in process time for solving
```

Figure 7: Time Breakdown

```
c
c seconds are measured in process time for solving
c
c --- [ glue usage ] -------------------------------------------------------
c
c focused glue 2 used 1099 clauses 29.42% accumulated 51.11% tier1
c focused glue 3 used 833 clauses 22.30% accumulated 73.41% tier2
c
c stable glue 2 used 681 clauses 31.09% accumulated 40.22% tier1
c stable glue 3 used 512 clauses 23.37% accumulated 63.60%
c stable glue 4 used 333 clauses 15.20% accumulated 78.81%
c stable glue 5 used 201 clauses 9.178% accumulated 87.99%
c stable glue 6 used 112 clauses 5.114% accumulated 93.10% tier2
c
c
c --- [ resources ] ---------------------------------------------------------
c
c total process time since initialization:         0.06    seconds
c total real time since initialization:            0.07    seconds
c maximum resident set size of process:            5.88    MB
c
c --- [ shutting down ] -----------------------------------------------------
c
c exit 10
```

Figure 8: Key Statistics

8.2 Conclusion Time

```
c
c seconds are measured in process time for solving
c
c --- [ glue usage ] -------------------------------------------------------
c
c focused glue 2 used 1099 clauses 29.42% accumulated 51.11% tier1
c focused glue 3 used 833 clauses 22.30% accumulated 73.41% tier2
c
c stable glue 2 used 681 clauses 31.09% accumulated 40.22% tier1
c stable glue 3 used 512 clauses 23.37% accumulated 63.60%
c stable glue 4 used 333 clauses 15.20% accumulated 78.81%
c stable glue 5 used 201 clauses 9.178% accumulated 87.99%
c stable glue 6 used 112 clauses 5.114% accumulated 93.10% tier2
c
c
c --- [ resources ] ---------------------------------------------------------
c
c total process time since initialization:         0.06    seconds
c total real time since initialization:            0.07    seconds
c maximum resident set size of process:            5.88    MB
c
c --- [ shutting down ] -----------------------------------------------------
c
c exit 10
```

Figure 9: Conclusion Time

9 Unsatisfiable Example

9.1 Start of Process

```
c --- [ solving ] ---------------------------------------------------------
c
c time measured in process time since initialization
c
c  seconds  reduction conflicts      size tier2     variables
c          MB    restarts   redundant    glue  trail      remaining
c           level      rate       size/glu tier1 irredundant
c
c 0  0.00  4  0 0   0  0    1    0 0.0 0 0 2 6  0%    3   0  0%
c
c --- [ result ] ------------------------------------------------------------
c
s UNSATISFIABLE
c
c ---[ run-time profiling ] -------------------------------------------------
c
c process time taken by individual solving procedures
c (percentage relative to process time for solving)
c
c          0.00  263.33% parse
c          0.00    0.00% search
c          0.00    0.00% simplify
c    ================================
c          0.00   14.14% solve
c
c last line shows process time for solving
c (percentage relative to total process time)
```

Figure 10: Start of UNSAT Process

9.2 UNSAT Verdict

```
c
c ---[ statistics ]---------------------------------------------------------
c
c conflicts:                 1       4166.67    per second
c fixed:                     3        100.00 %  of all variables
c learned:                   0          0.00 %  per conflict
c minimized:                 0          0.00 %  learned literals
c shrunken:                  0          0.00 %  learned literals
c minishrunken:              0          0.00 %  learned literals
c otfs:                      0          0.00 %  of conflict
c propagations:              2          0.01 M  per second
c ticks:                     6          3.00    propagation
c
c seconds are measured in process time for solving
c
```

Figure 11: Non-Satisfiability Verdict

9.3 Statistics

```
c seconds are measured in process time for solving
c
c --- [ glue usage ] ---------------------------------------------------------
c
c focused glue 2 used 0 clauses 0.000% accumulated 0.000% tier1
c focused glue 3 used 0 clauses 0.000% accumulated 0.000%
c focused glue 4 used 0 clauses 0.000% accumulated 0.000%
c focused glue 5 used 0 clauses 0.000% accumulated 0.000%
c focused glue 6 used 0 clauses 0.000% accumulated 0.000% tier2
c
c stable glue 2 used 0 clauses 0.000% accumulated 0.000% tier1
c stable glue 3 used 0 clauses 0.000% accumulated 0.000%
c stable glue 4 used 0 clauses 0.000% accumulated 0.000%
c stable glue 5 used 0 clauses 0.000% accumulated 0.000%
c stable glue 6 used 0 clauses 0.000% accumulated 0.000% tier2
c
c
```

Figure 12: UNSAT Key Statistics

9.4 Summary and Exit Information

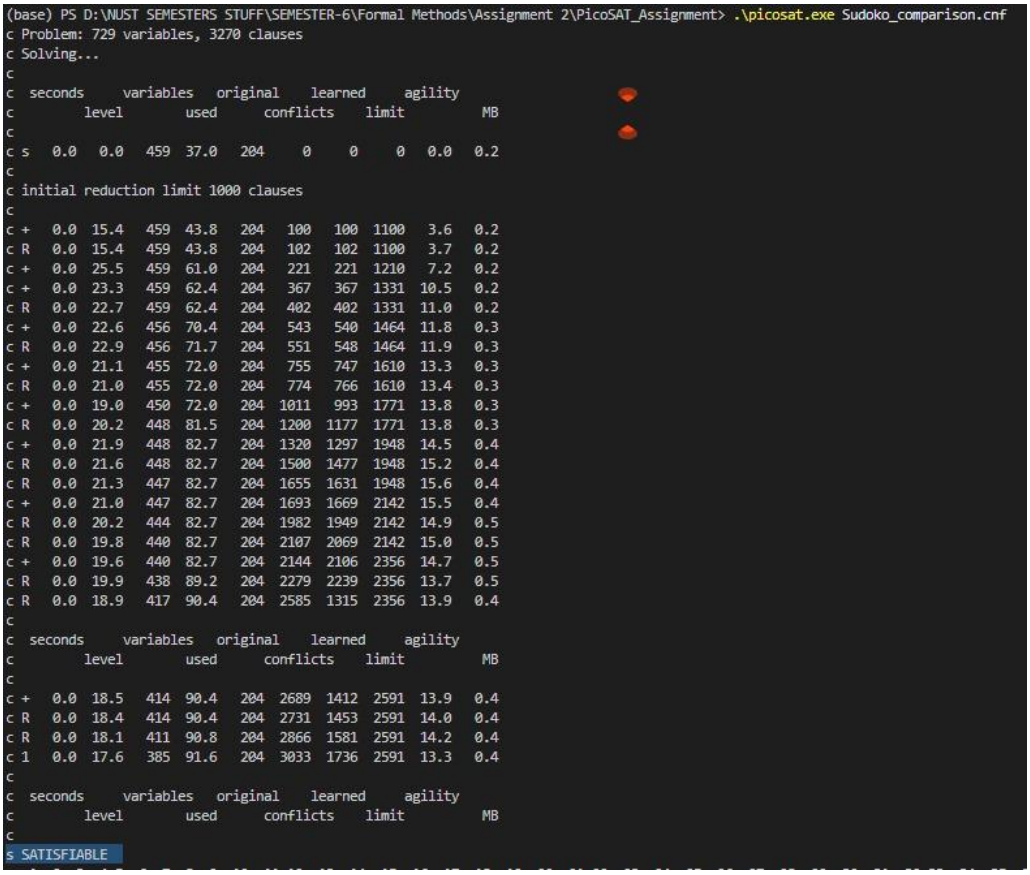
```
c
c
c --- [ resources ] ------------------------------------------------------------
c
c total process time since initialization:      0.00     seconds
c total real time since initialization:         0.00     seconds
c maximum resident set size of process:         4.20     MB
c
c --- [ shutting down ] -------------------------------------------------------
c
c exit 20

┌──(faisaliqbal㉿kali)-[~/fm-sat/cadical]
└─$ 
```

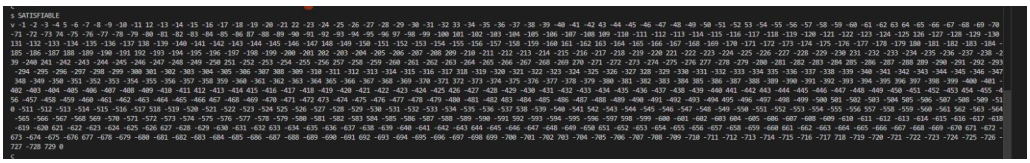
Figure 13: Process Exit Summary

10 Comparison with PicoSAT

Comparison with Hassan Irfan's group using PicoSAT can be included here, analyzing:



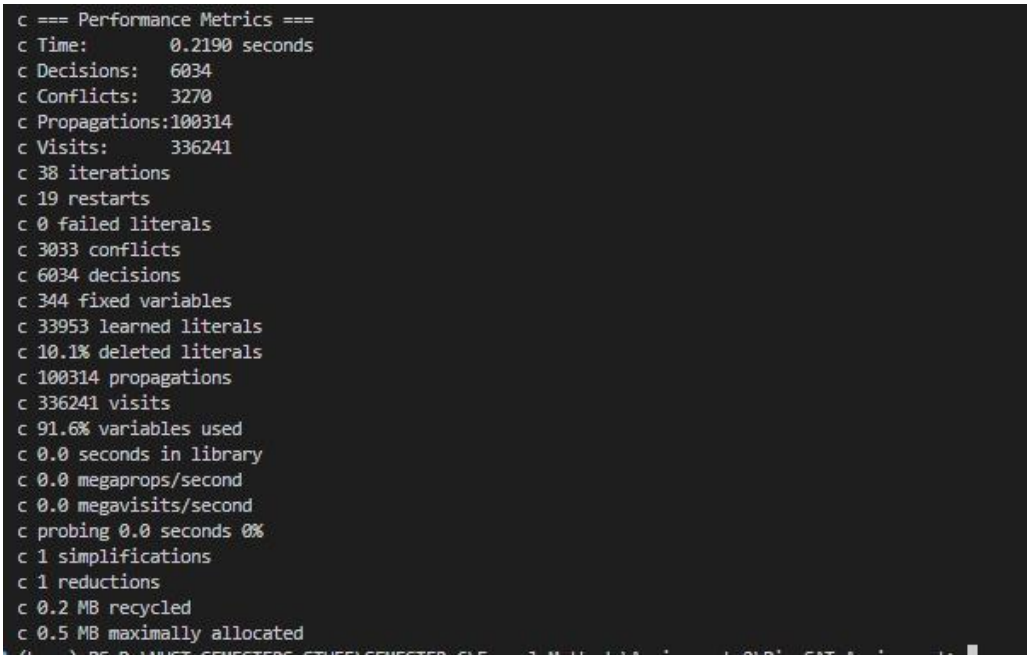
```
(base) PS D:\NUST SEMESTERS STUFF\SEMESTER-6\Formal Methods\Assignment 2\PicoSAT_Assignment> .\picosat.exe Sudoko_comparison.cnf
c Problem: 729 variables, 3270 clauses
c Solving...
c
c  seconds     variables   original    learned     agility
c         level        used      conflicts    limit          MB
c
c s   0.0   0.0   459  37.0   204     0     0     0   0.0   0.2
c
c initial reduction limit 1000 clauses
c
c +   0.0  15.4   459  43.8   204   100   100  1100   3.6   0.2
c R   0.0  15.4   459  43.8   204   102   102  1100   3.7   0.2
c +   0.0  25.5   459  61.0   204   221   221  1210   7.2   0.2
c +   0.0  23.3   459  62.4   204   367   367  1331  10.5   0.2
c R   0.0  22.7   459  62.4   204   402   402  1331  11.0   0.2
c +   0.0  22.6   456  70.4   204   543   540  1464  11.8   0.3
c R   0.0  22.9   456  71.7   204   551   548  1464  11.9   0.3
c +   0.0  21.1   455  72.0   204   755   747  1610  13.3   0.3
c R   0.0  21.0   455  72.0   204   774   766  1610  13.4   0.3
c +   0.0  19.0   450  72.0   204  1011   993  1771  13.8   0.3
c R   0.0  20.2   448  81.5   204  1200  1177  1771  13.8   0.3
c +   0.0  21.9   448  82.7   204  1320  1297  1948  14.5   0.4
c R   0.0  21.6   448  82.7   204  1500  1477  1948  15.2   0.4
c R   0.0  21.3   447  82.7   204  1655  1631  1948  15.6   0.4
c +   0.0  21.0   447  82.7   204  1693  1669  2142  15.5   0.4
c R   0.0  20.2   444  82.7   204  1982  1949  2142  14.9   0.5
c R   0.0  19.8   440  82.7   204  2107  2069  2142  15.0   0.5
c +   0.0  19.6   440  82.7   204  2144  2106  2356  14.7   0.5
c R   0.0  19.9   438  89.2   204  2279  2239  2356  13.7   0.5
c R   0.0  18.9   417  90.4   204  2585  1315  2356  13.9   0.4
c
c  seconds     variables   original    learned     agility
c         level        used      conflicts    limit          MB
c
c +   0.0  18.5   414  90.4   204  2689  1412  2591  13.9   0.4
c R   0.0  18.4   414  90.4   204  2731  1453  2591  14.0   0.4
c R   0.0  18.1   411  90.8   204  2866  1581  2591  14.2   0.4
c 1   0.0  17.6   385  91.6   204  3033  1736  2591  13.3   0.4
c
c  seconds     variables   original    learned     agility
c         level        used      conflicts    limit          MB
c
s SATISFIABLE
```



```
s SATISFIABLE
v -1 -2 -3 -4 5 -6 -7 -8 -9 -10 -11 12 -13 -14 -15 -16 -17 -18 -19 -20 -21 22 -23 -24 -25 -26 -27 -28 -29 -30 -31 -32 33 -34 -35 -36 -37 -38 -39 -40 -41 -42 43 -44 -45 -46 -47 -48 -49 -50 -51 -52 53 -54 -55 -56 -57 -58 -59 -60 -61 -62 63 64 -65 -66 -67 -68 -69 -70
-71 -72 -73 74 -75 -76 -77 -78 -79 -80 -81 -82 -83 -84 -85 -86 87 -88 -89 -90 -91 -92 -93 -94 -95 -96 97 -98 -99 -100 101 -102 -103 -104 -105 -106 -107 -108 109 -110 -111 -112 -113 -114 -115 -116 -117 -118 -119 -120 -121 -122 -123 -124 -125 126 -127 -128 -129 -130
131 -132 -133 -134 -135 -136 -137 138 -139 -140 -141 -142 -143 -144 -145 -146 -147 148 -149 -150 -151 -152 -153 -154 -155 -156 -157 -158 -159 -160 161 -162 163 -164 -165 -166 -167 -168 -169 -170 -171 -172 -173 -174 -175 -176 -177 -178 -179 180 -181 -182 -183 -184 -
185 -186 -187 188 -189 -190 -191 192 -193 -194 -195 -196 -197 -198 -199 -200 -201 202 -203 -204 -205 -206 -207 -208 209 -210 -211 -212 -213 -214 -215 -216 -217 -218 -219 -220 221 -222 -223 -224 -225 -226 -227 -228 -229 -230 231 -232 -233 -234 -235 -236 -237 -238 -2
39 -240 241 -242 -243 -244 -245 -246 -247 -248 -249 -250 251 -252 -253 -254 -255 -256 257 -258 -259 -260 -261 -262 -263 -264 -265 -266 -267 -268 -269 270 -271 -272 -273 -274 -275 -276 277 -278 -279 -280 -281 -282 -283 -284 285 -286 -287 -288 289 -290 -291 -292 -293
-294 -295 -296 -297 -298 -299 -300 301 -302 -303 -304 -305 -306 -307 308 -309 -310 -311 -312 -313 -314 -315 -316 -317 318 -319 -320 -321 -322 -323 -324 -325 -326 -327 328 -329 -330 -331 -332 -333 -334 335 -336 -337 -338 -339 -340 -341 -342 -343 -344 -345 -346 -347
348 -349 -350 -351 -352 -353 -354 -355 -356 -357 -358 359 -360 -361 362 -363 -364 365 -366 -367 -368 -369 -370 -371 372 -373 -374 -375 -376 -377 -378 -379 -380 -381 -382 -383 -384 385 -386 -387 -388 -389 -390 -391 -392 -393 -394 -395 396 397 -398 -399 -400 -401 -
402 -403 -404 -405 -406 -407 -408 -409 -410 -411 412 -413 -414 415 -416 -417 -418 -419 -420 -421 -422 -423 -424 -425 426 -427 -428 -429 -430 -431 -432 -433 -434 -435 -436 -437 -438 -439 -440 441 -442 443 -444 -445 -446 -447 -448 -449 -450 -451 -452 -453 454 -455 -4
56 -457 -458 -459 -460 -461 -462 -463 -464 -465 -466 467 -468 -469 -470 -471 -472 473 -474 -475 -476 -477 -478 -479 -480 -481 -482 483 -484 -485 -486 -487 -488 -489 -490 -491 -492 -493 -494 495 -496 -497 -498 -499 -500 501 -502 -503 -504 505 -506 -507 -508 -509 -51
0 -511 -512 -513 -514 -515 -516 -517 518 -519 -520 -521 -522 -523 -524 525 -526 -527 -528 -529 -530 -531 -532 -533 -534 -535 -536 -537 538 -539 -540 -541 542 -543 -544 -545 -546 -547 -548 -549 -550 -551 -552 -553 -554 -555 -556 557 -558 -559 -560 -561 562 -563 -564
-565 -566 -567 -568 569 -570 -571 -572 -573 -574 -575 -576 -577 -578 -579 -580 -581 -582 -583 584 -585 -586 -587 -588 -589 -590 -591 592 -593 -594 -595 -596 -597 598 -599 -600 -601 -602 -603 604 -605 -606 -607 -608 -609 -610 -611 -612 -613 -614 -615 -616 -617 -618
-619 -620 621 -622 -623 -624 -625 -626 627 -628 -629 -630 -631 -632 633 -634 -635 -636 -637 -638 -639 -640 -641 -642 -643 644 -645 -646 -647 -648 -649 -650 651 -652 -653 -654 -655 -656 -657 -658 -659 -660 661 -662 -663 -664 -665 -666 -667 -668 -669 -670 671 -672 -
673 -674 -675 -676 677 -678 -679 -680 -681 -682 -683 -684 -685 -686 -687 -688 -689 -690 -691 692 -693 -694 -695 -696 -697 -698 699 -700 -701 -702 703 -704 -705 -706 -707 -708 -709 -710 -711 -712 -713 -714 -715 -716 -717 718 -719 -720 -721 -722 -723 -724 -725 -726 -
727 -728 729 0
```

Figure 14: PicoSAT

11 Overall performance



```
c === Performance Metrics ===
c Time:        0.2190 seconds
c Decisions:   6034
c Conflicts:   3270
c Propagations:100314
c Visits:      336241
c 38 iterations
c 19 restarts
c 0 failed literals
c 3033 conflicts
c 6034 decisions
c 344 fixed variables
c 33953 learned literals
c 10.1% deleted literals
c 100314 propagations
c 336241 visits
c 91.6% variables used
c 0.0 seconds in library
c 0.0 megaprops/second
c 0.0 megavisits/second
c probing 0.0 seconds 0%
c 1 simplifications
c 1 reductions
c 0.2 MB recycled
c 0.5 MB maximally allocated
```

Figure 15: Performance Metric